

# The ICT Lounge

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## Section 7.6: Modelling Applications

Computers can be used to **simulate different scenarios** to work out all of the possible outcomes. This is called '**modelling**'.

Examples of computer modelling include:

- **Flight simulators to help train aircraft pilots**
- **3D models of buildings and products to simulate how designs will look when built**
- **Spreadsheet models to simulate profit/loss based on different levels of income.**

### Key Concepts of this section:

- # Understand how computers can be used to '**model**' or '**simulate**' real-life situations.
- # Know some **advantages / disadvantages** of computer modelling.
- # Be able to discuss different **examples of computer modelling**.

## Computer Modelling and Simulation

### What is a computer model/simulation?

- # Simulations aim to mimic (copy) real life systems in order to see the outcomes of different scenarios.
- # Another name for a computer simulation that mimics real-life situations is a '**computer model**'.
- # Computer simulations can be created for a range of applications including:

[Finance Simulation](#)[Flight Simulation](#)[Driving Simulation](#)[3D Models](#)

### Key Words:

Modelling, Simulation, Prediction, Training, Design.

### What are computer models used for?

#### Prediction

Models can predict outcomes based on inputs we enter (e.g. spreadsheets)

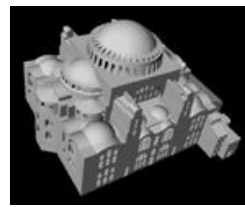
#### Training

For example, allowing pilots or surgeons to learn their skills safely.

#### Design

Architects can produce very accurate

| Profit/Loss                         |
|-------------------------------------|
| = $(E3 \times F3) - (C3 \times D3)$ |
| = $(E4 \times F4) - (C4 \times D4)$ |
| = $(E5 \times F5) - (C5 \times D5)$ |
| = $(E6 \times F6) - (C6 \times D6)$ |
| = $(E7 \times F7) - (C7 \times D7)$ |
| =SUM(G3:G7)                         |



3D models of buildings.

## Advantages of using computer models/simulations

# Using computer simulations and models to mimic real-life situations can bring many advantages. Some include:

### Cheaper -



Modelling a situation is often much cheaper than carrying out the task for real. For example, architects can **make and solve design errors on simulated buildings** rather than waiting until the real thing is constructed before realising.

### Safer -



Flight simulators mimic the skills needed to fly a plane. Novice pilots are **likely to crash the plane** whilst they are **learning**. This could be **fatal if piloting a real aircraft**. If a simulated plane is crashed, they just **reload the simulation** and try again.

### Saves Time -



Simulations can produce results **faster than the real thing**. For example, **global warming** models can predict the **temperature of the earth in 50 years time** without actually having to wait that long.

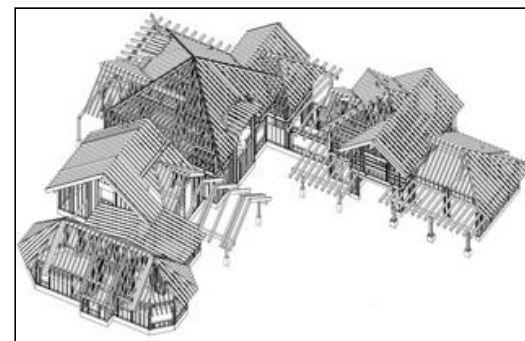
### Greater Range of Situations -



Simulations can be programmed to mimic a wide range of **extreme situations**. For example, **pilots** can be exposed to **storms, volcanic ash clouds, high winds** etc. In real life these conditions cannot be guaranteed.



Crashing a plane in a flight-sim is safe.



Buildings can be perfected in a 3D model before money is spent on actual construction

## Disadvantages of using computer models/simulations

# Disadvantages of using computer simulations/models include:

### Accuracy -

Modelling a situation can **never perfectly mimic real life**.

**Initial  
Expense -**

→ There will always be some **difference** between the **model and reality**.

→ Simulators require **sophisticated hardware and software**. For example, high powered processors and lots of RAM. The simulation software (program) can also be expensive.

**Programmer  
Error -**

→ If the simulation's **programmer has made errors** whilst creating the program, the **simulation outcomes** will be more **inaccurate**.



Some simulators require expensive hardware.

## Examples of Computer Models/Simulations

The following examples discuss some computer models:

### 1 Flight Simulator (used to train pilots)

# These mimic the **skills needed to pilot a real aircraft**. there are two types of flight simulator:

- **PC/Laptop simulation designed to be used at home**
- **Replica cockpit simulations which are large and expensive but feel more like the real thing.**

# The simulator allows the pilot to practice the following flight skills:

- **Use of cockpit controls**
- **Take off / Land**
- **Navigation (following correct path to a destination)**
- **Handling different conditions (high wind, storms, engine failure)**



Cockpit flight sim with realistic controls

- # At first trainee pilots will **make mistakes** and maybe even **crash the plane**. This doesn't matter because they can simply **reset the simulation** and try again. Nobody gets hurt.
- # When the trainee **gains enough experience** in the simulator, they will be allowed to **fly a real plane**.

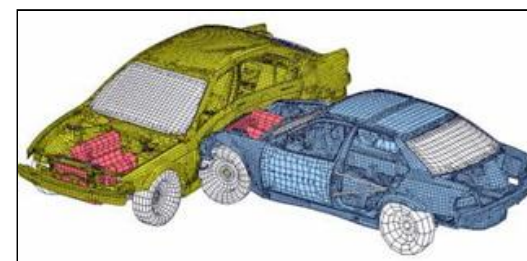
**PLAY VIDEO TO  
SEE AN EXAMPLE  
OF A COCKPIT  
FLIGHT SIM**



PC flight sim for use in the home

## 2 Driving Simulation (entertainment, training and crash testing)

- # These mimic the skills required to **control vehicles** such as **cars, trucks, buses** and **trains**.
- # Uses of driving simulators as a training tool include:
  - **Use of vehicle controls (steering wheel, accelerator, gears, clutch etc)**
  - **Practice driving maneuvers (emergency stop, reversing, parking etc)**
  - **Learning how to handle large trucks safely**
  - **Driving in hazardous conditions (snow, heavy rain etc)**
- # As with flight simulators, trainee drivers can build up a **high level of confidence** using the driving simulation **before they actually begin to drive for real**. Any mistakes or crashes are just '**virtual**' and vehicles are **not damaged** or any **people hurt**.
- # Some driving simulators are used to mimic the **effects of high-speed car crashes** on the driver and passengers.



Crash-Test simulators allow engineers to design and build safer cars

This replaces the old method of using **actual cars** and **crash-test dummies** and is much

**cheaper** and **easier to set up**.

Data taken from the simulation allows engineers to design and **build safer cars**. This is known as '**Computer Aided Engineering**' (CAE).

**PLAY VIDEO TO SEE AN  
EXAMPLE OF A TRUCK  
DRIVER TRAINING  
SIMULATOR**



Old methods of crash-testing required real cars and crash test dummies. This is much more expensive.



Some driving simulators are for entertainment purposes like Test Drive Unlimited 2

### 3 3D Models (buildings, rooms, mobile phones etc)

# 3D models are created using special software called '**Computer Aided Design**' (CAD).

CAD software can be used to create **computer generated prototypes** for a wide range of models including:

- **Buildings**
- **Rooms**
- **Products and gadgets (phones, mp3 players etc)**

#### **NOTE!**

A prototype is an **early stage design** for the intended product.

# 3D models are **created on the computer** and then **shown to the customer** so they can give the designer **feedback**. The designer then **applies changes** directly **to the model**.

# Creating and viewing 3D models on the computer offers advantages including:

3D models can be rotated to see all angles

**Zoom -** Images can be zoomed into so that fine detail can be seen clearly

**Rotation -** Images can be rotated through many different angles so that every aspect of the model can be seen

**Cheaper -** Creating and modifying prototypes on software does not require expensive, physical materials (like plastic, wood, bricks etc)

**Quicker -** CAD software allows prototypes to be created and modified by manipulating elements on a screen. This is much quicker than having to assemble and change plastic/wood based prototypes.

- # Once the **customer is happy** with the prototype model, the **product is manufactured** using **real materials**. Everyone can be sure that the final product is to an **exact specification** with no errors because the **3D model was perfected prior to production**. This can save lots and lots of money.

### NOTE!

Imagine manufacturing the actual products and then realising there were **design errors**. This would be a **hugely expensive problem to fix!**

3D computer prototype of a room design

3D model of a torch design. Note how the model demonstrates how the torch is assembled

## 4 Finance Simulation (business spreadsheet models)

- # Spreadsheet models can be used to **predict finance type outcomes** based on **mathematical values**.

These predictions can show us things like.

- **Profit / Loss**
- **Over budget / Within Budget**
- **Answers to 'What If' questions (see below)**

- # Spreadsheet models can be used to answer **'What If' questions**. For example - **"What if I give my staff a 10% pay rise. Will I still make a profit?"**

This is made possible with the use of **'formulae'** which recalculate results based on new data entered.

- # Running questions like this through the spreadsheet can help managers make **'informed decisions'** and **reduce** the risk of **overspending and reduced profits**.

**Picture Examples:**



Spreadsheet programs such as Microsoft Excel are commonly used to create finance models

Salary Increase: 10%

| Employee            | Salary  | Salary with increase |
|---------------------|---------|----------------------|
| Michael             | £30,000 | £33,000              |
| Tanja               | £20,000 | £22,000              |
| Grant               | £25,000 | £27,500              |
| Total Salary Costs: |         | £82,500              |
| Gross Profit:       |         | £100,000             |
| Net Profit:         |         | £17,500              |

Spreadsheet model showing what net profit is if employees receive a 10% pay increase.

Salary Increase: 40%

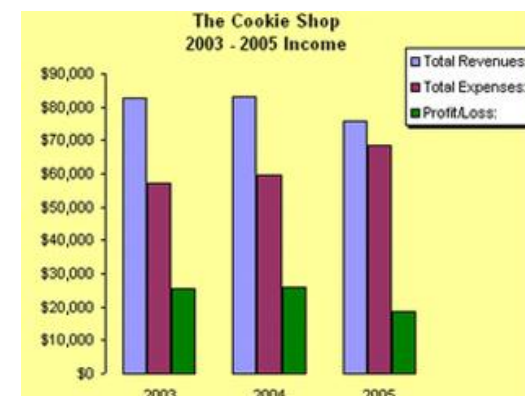
| Employee            | Salary  | Salary with increase |
|---------------------|---------|----------------------|
| Michael             | £30,000 | £42,000              |
| Tanja               | £20,000 | £28,000              |
| Grant               | £25,000 | £35,000              |
| Total Salary Costs: |         | £105,000             |
| Gross Profit:       |         | £100,000             |
| Net Profit:         |         | −£5,000              |

By increasing the employee pay by 40%, the model predicts that the business will lose money.

Salary Increase: 0.4

| Employee            | Salary | Salary with increase   |
|---------------------|--------|------------------------|
| Michael             | 30000  | $= (D5 * \$D\$2) + D5$ |
| Tanja               | 20000  | $= (D6 * \$D\$2) + D6$ |
| Grant               | 25000  | $= (D7 * \$D\$2) + D7$ |
| Total Salary Costs: |        | $= \text{SUM}(E5:E7)$  |
| Gross Profit:       |        | 100000                 |
| Net Profit:         |        | $= E9 - E8$            |

Spreadsheet models use formulae to calculate their predictions.



Spreadsheets also make use of charts/graphs to show changes to the model

### NOTE!

Unit 14: Data Analysis of the course syllabus is all about spreadsheet modelling.

**Previous - Control Applications - Examples**

### Links to Theory Units:

[Section 1: Types and Components of Computer Systems](#)

[Section 4: Computer Networks](#)

[Section 7: The Ways in which ICT is used](#)

[Section 2: Input and Output Devices](#)

[Section 5: Data Types](#)

[Section 7: Part 2](#)

[Section 3: Storage Devices and Media](#)

[Section 6: The Effects of Using ICT](#)

[Section 8: Systems Analysis and Design](#)

## Links to Practical Units:

[Section 9: Communication](#)

Section 12: Integration

[Section 15: Web site Authoring](#)

[Section 10: Document Production](#)

Section 13: Output Data

[Section 16: Presentation Authoring](#)

[Section 11: Data Manipulation](#)

[Section 14: Data Analysis](#)

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